AL-HIKMA POLYTECHNIC

STUDENT MANAGEMENT SYSTEM (SMS) ↔ LEARNING MANAGEMENT SYSTEM (LMS)

TECHNICAL INTEGRATION & API CONTRACT

Version 2.0

Purpose. This document defines the agreed integration between the Student Management System (SMS/Portal) and the Learning Management System (LMS). It establishes which system owns each category of data, how students access the LMS, and how student, course, enrollment and assessment information is synchronized.

1. Purpose

The SMS/Portal is the authoritative source of truth for student identity, authentication, academic structure, course registration, enrollment, student status and official academic results.

The LMS is the online learning platform responsible for delivering online learning content, activities, assessments, submissions and learning progress.

The LMS may maintain local copies of SMS data where required for normal operation, but those copies are not authoritative.

The integration must support both regular/onsite students, who primarily use the Portal, and online students, who use the LMS for online learning.

2. System Ownership

2.1 SMS / Portal — System of Record

• Student identity and matriculation number

• Student authentication

• Student name and email

• Department and programme

• Academic level, session and semester

• Student status

• Student delivery/study mode

• Courses and course definitions

• Course registration/enrollment

• Official academic results and grades

• Academic progression and other institutional student-management information

2.2 LMS — Online Learning System

• Online course content, lessons, modules, videos and materials

• Online assignments, quizzes and examinations

• Assignment submissions and assessment attempts

• Learning progress and lesson completion

• Online learning activity and discussions

• LMS-specific notifications

• Online assessment results before official academic processing

3. Core Architectural Principle

SMS / Portal = Identity + Academic System of Record
The Portal owns authoritative student and academic information.

LMS = Online Learning Delivery System
The LMS consumes the required academic information and manages the online learning experience.

4. Student Identity

• `matric_number` is the permanent integration key between the SMS and LMS.

• `matric_number` must be unique in the SMS and must not change during the student's lifetime.

• The LMS may have its own internal database ID, but that ID must not replace `matric_number` as the integration key.

Example

```
SMS student: internal ID = 8f3..., matric_number = CS/ND1/001
LMS student: internal ID = 91a..., matric_number = CS/ND1/001

Integration relationship: SMS.matric_number = LMS.matric_number
```

5. Student Authentication and Single Sign-On

• Students authenticate through the SMS/Portal.

• The LMS must not maintain independent student passwords or student registration.

• An authenticated Portal session is used to initiate LMS access through SSO.

• The Portal generates a short-lived, signed authentication assertion and redirects the browser to the LMS.

• The LMS validates the assertion, identifies the student, checks LMS eligibility, creates its own LMS session and redirects the student to the LMS dashboard.

• The SMS session cookie must not be shared with the LMS, and the LMS must not access the SMS session store directly.

5.1 SSO Flow

```
Student
↓
Authenticated Portal session
↓
Portal verifies LMS eligibility
↓
Portal generates short-lived signed assertion
↓
LMS validates assertion
↓
LMS finds student by matric_number
↓
LMS creates its own session
↓
LMS dashboard
```

5.2 SSO Token Requirements

• Recommended token lifetime: 30–60 seconds.

• Use digital signing; asymmetric signing is preferred.

• The SMS keeps the private signing key; the LMS receives the public verification key.

• The LMS must validate issuer, audience, expiry and the student's identifier.

• Replay protection should be implemented where practical.

• The assertion is only for establishing the LMS session; it is not the permanent LMS session.

6. Server-to-Server Authentication

The existing API-token approach remains for SMS-to-LMS synchronization:

```
Authorization: Bearer <LMS_API_TOKEN>
Content-Type: application/json
```

• The token is server-side only.

• It must never be exposed to browser clients or frontend JavaScript.

• It must be stored securely and be rotatable/revocable.

• All production communication must use HTTPS.

7. Student Synchronization

The LMS will expose an idempotent student synchronization endpoint:

```
POST /api/v1/sync/students
```

```
{
"matric_number": "CS/ND1/001",
"first_name": "Amina",
"last_name": "Bello",
"email": "amina@example.com",
"department_code": "CS",
"academic_level": "ND1",
"current_session": "2025/2026",
"semester": "2",
"status": "active",
"lms_access": true
}
```

• If the student exists, update the local LMS representation.

• If the student does not exist, create the local LMS representation.

• Repeated requests must not create duplicates.

• The LMS must not treat its local copy as more authoritative than the SMS.

8. LMS Access

• The SMS determines whether a student is eligible to use the LMS.

• A boolean `lms_access` field is recommended so the LMS is not tied to a specific delivery-mode enum.

• If `lms_access` becomes false, the LMS must disable access.

• During SSO, the LMS must ensure the student is still eligible to access the LMS.

9. Courses and Course Ownership

• The SMS is authoritative for academic courses, including course code, title, credit unit, academic level, session, semester and department relationship.

• The LMS may maintain local course records because online content needs to be attached to courses.

• An LMS course record is a projection of the corresponding SMS course.

• The LMS may add LMS-specific information such as learning modules, videos, assignments, quizzes and other content.

• The LMS must not independently change the authoritative academic identity of a course.

10. Course Enrollment

Student course registration is authoritative in the SMS.

```
{
"matric_number": "CS/ND1/001",
"session": "2025/2026",
"semester": "2",
"course_codes": ["COM101", "MTH101", "BUS101"]
}
```

• The LMS should make online learning spaces available according to the student's SMS registration.

• The LMS should not independently determine official student registration from department + level + semester when authoritative registration data is available from the SMS.

• Temporary LMS-derived enrollment may be used during initial setup if required, but SMS-driven enrollment is the target architecture.

11. Online Learning Content

• The LMS owns the online representation of a course.

• This includes modules, lessons, videos, reading materials, assignments, quizzes, online examinations and related learning activities.

• The SMS does not need to manage these LMS-specific learning assets.

12. Results and Assessments

• The LMS may store online assessment attempts and scores such as quizzes, assignments and online examinations.

• The LMS is not the authoritative system for the student's final academic result.

• The official academic result remains in the SMS.

• Where required, the LMS will send assessment results to the SMS for official academic processing.

```
POST /api/v1/sync/assessment-results

{ "matric_number": "CS/ND1/001", "course_code": "COM101", "assessment_type": "quiz", "score": 17,
"maximum_score": 20, "session": "2025/2026", "semester": "2" }
```

13. Regular and Online Students

Both regular and online students remain students of the SMS.

Regular student: uses the Portal for normal academic activities and does not necessarily require LMS access.
Online student: uses the LMS for online learning while identity, enrollment and official academic records remain in the SMS.

14. Data Synchronization Model

14.1 Initial synchronization

A one-time full synchronization will be performed when the LMS is first connected. The existing student endpoint may support the initial load and incremental synchronization.

14.2 Real-time synchronization

• Student created or updated

• Student status changed

• LMS access changed

• Course created or updated

• Student course registration changed

14.3 Nightly reconciliation

A nightly reconciliation process should compare the relevant SMS data with the LMS projection and repair missed or failed synchronization events.

15. Asynchronous Synchronization

SMS database operations must not depend on the LMS being available.

```
Portal request
↓
Save SMS database
↓
Commit transaction
↓
Queue synchronization job
↓
Return response
↓
Worker synchronizes with LMS
```

• Synchronization should be handled asynchronously using the existing queue infrastructure where appropriate.

• Temporary LMS outages should cause retryable synchronization failures, not failure of the underlying SMS operation.

• Failed jobs should be visible and retryable.

16. Queue Payloads

Where possible, queue jobs should contain the SMS record identifier rather than a complete snapshot of student data. The worker should fetch the current record before sending it to the LMS.

```
{ "student_id": "internal-sms-id" }
```

17. Error Handling

```
400 Bad Request — invalid request
401 Unauthorized — missing/invalid API token
403 Forbidden — authenticated but not permitted
404 Not Found — resource not found
409 Conflict — conflicting state where applicable
422 Unprocessable Entity — valid JSON but invalid business rule
500 Server Error — internal LMS error
```

18. Security Requirements

• All production communication must use HTTPS.

• API tokens and signing keys must remain server-side.

• SSO assertions must be short-lived and cryptographically verified.

• Expired or invalid SSO assertions must be rejected.

• Authentication endpoints should be rate-limited.

• Sensitive credentials and tokens must never be written to logs.

• Browser-supplied student information must not be treated as authoritative.

19. Auditability

Both systems should log important integration events, including student synchronization, synchronization failures, course/enrollment synchronization, successful and failed SSO, LMS access changes and assessment-result submissions. Logs should include useful identifiers and timestamps without recording secrets.

20. API Ownership

SMS team

• Student, course and enrollment source data

• Student authentication

• SSO assertion issuance

• Official academic results

• LMS access eligibility

LMS team

• Synchronization endpoints

• Idempotent upsert logic

• Local student/course projections

• Online learning content

• Online activities and assessments

• Learning progress

• LMS session management

• SSO assertion validation

21. Source-of-Truth Rule

When data conflicts between the two systems, the SMS wins for all student and academic-management information.

For example, if the SMS says a student is registered for COM101 and MTH101 while the LMS has COM101 and BUS101, the LMS must reconcile its local enrollment against the SMS rather than changing the SMS data.

22. Important Note on Local LMS Tables

The LMS may and should have local Student and Course tables where they are needed for the LMS application. Having these tables does not mean the LMS becomes a second source of truth. They are local projections of data owned by the Portal and should be updated from the Portal through the agreed synchronization mechanisms. The LMS remains responsible for its own learning-specific data—such as lessons, videos, assignments, submissions, quizzes, learning activity and progress—which does not need to be maintained by the Portal.

23. SSO User Flow

1. Student logs into the Portal.

2. Portal creates the normal Portal session.

3. Student selects 'Open LMS'.

4. Portal verifies the session and LMS eligibility.

5. Portal creates a short-lived signed SSO assertion.

6. Browser is redirected to the LMS SSO endpoint.

7. LMS validates the assertion.

8. LMS identifies the student using matric_number.

9. LMS creates its own LMS session.

10. Student enters the LMS without a second login.

24. Non-Requirements

• Independent LMS student registration

• Independent LMS student passwords

• Independent LMS password reset for students

• Independent official academic course registration

• Independent official academic result management

• Independent student identity management

25. Implementation Priority

Phase 1 — Data foundation

• Student synchronization

• Course synchronization

• Student-course enrollment synchronization

• LMS access flag

• Idempotent upsert logic

• Initial synchronization

• Nightly reconciliation

Phase 2 — Authentication

• SMS SSO endpoint

• Signed short-lived assertion

• LMS SSO endpoint

• Assertion verification

• LMS session creation

• LMS eligibility checks

Phase 3 — Online learning

• Course content

• Lessons

• Videos

• Materials

• Assignments

• Quizzes

• Online assessments

• Learning progress

Phase 4 — Academic integration

• Online assessment-result synchronization

• SMS result integration

• Official result processing

26. Acceptance Criteria

1. A student can log into the Portal and access the LMS without a second student password.
2. The LMS cannot create an independent student identity without an SMS-originated student record.
3. Student identity is synchronized from the SMS.
4. Student course registration is synchronized from the SMS.
5. The LMS cannot override authoritative SMS academic information.
6. Regular students can remain Portal-only users.
7. Online students can access the LMS through SSO.
8. Disabling LMS access in the SMS prevents future LMS SSO access.
9. SMS operations do not fail because the LMS is temporarily unavailable.
10. Failed synchronization jobs can be retried.
11. Repeated synchronization requests do not create duplicate LMS records.
12. Nightly reconciliation can repair missed synchronization events.
13. LMS assessment results can be transmitted to the SMS where required.
14. Official academic results remain controlled by the SMS.
15. No LMS API credentials are exposed to browser clients.
16. SSO assertions are short-lived and cryptographically verified.
17. Integration events can be audited and traced.

> Final architectural position: The SMS/Portal is the institution's identity and academic system of record. The LMS is the institution's online learning system. The LMS may maintain local Student and Course tables as projections of Portal data, while all LMS-specific learning content and activity remain owned by the LMS.

Document status: Proposed technical integration contract for implementation agreement.